MIL-T-8249B (USAF)
26 JUNE 1979
SUPERSEDING
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10 July 1956

MILITARY SPECIFICATION

TRANSMITTER, RATE OF FLOW, FUEL, TYPE MB-2

This specification is approved for use by Code 99, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of remote indicating, fuel, rate of flow transmitter, designated type MB-2.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or requests for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

FEDERAL

PPP-B-636	- Box, Shipping, Fiberboard

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of
MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-F-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-I-7566	- Indicator System, Fuel Flowmeter, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq AFLC CASO/LODS, Federal Center, Battle Creek, MI 49016, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6620

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MIL-P-8422 - Power Supply, 28V DC Input, 15V 4-Cycle 3-Phase output, Type MC-1

STANDARDS

MILITARY

MIL-STD-129 - Marking of Shipments and Storage
MIL-STD-130 - Identification Marking of U.S. Military Property
MIL-STD-794 - Parts and Equipment, Procedures for Packaging and Packing of
MS33649 - Boss, Fluid Connection - Internal Straight Thread
MS33678 - Connector, Receptacle, Electric, Integral Mounting

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. This specification makes provisions for first article testing (see 4.1 and 6.2).

3.2 General specification. The requirements of MIL-I-7566 apply as requirements of this specification with the exceptions and additions called out herein. When the two specifications conflict, this specification shall govern.

3.3 Design and construction.

3.3.1 General. The transmitter shall be designed and calibrated to operate satisfactorily with a 0-12,000-pph indicator in accordance with General Electric part number 8DJ64GBH1 or equivalent, and power supply in accordance with MI-P-8422.

3.3.1.1 The transmitter remote indication system shall operate on the second harmonic system in which the moving element consists of a permanent 2-pole magnet, and the toroidally wound stator is energized with 115V 400-cycle single phase ac.

3.3.2 Range. The transmitter shall measure flow rated from 0-12,000 pounds per hour.

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3.3.2.1 Mechanical stop. A mechanical stop shall be provided within the transmitter at a minimum of 13,000 pph flow rate but not more than a 13,500 pph flow point.

3.3.3 Operating pressure. The high pressure transmitter shall operate under a pressure of 600 psi and withstand a pressure of 1600 psi without damage.

3.3.4 Pressure drop. The transmitter, by virtue of its design, shall not have appreciably more pressure drop under any fuel flow conditions when inoperative than when it is operating normally.

3.3.5 Temperature range. The transmitter shall be designed to withstand an ambient operating temperature range of -55°C to +121°C with fuel flowing through the transmitter, and exposure to +190°C with no fuel flowing through the transmitter.

3.3.6 Electrical receptacles. The transmitter's electrical receptacles shall be in accordance with MS33678-14S-7P for the operating power source, and MS33678-14S-2P for connection to the flow rate indicator in accordance with figure 1.

3.3.7 Power consumption. The power consumption of the pick-off coil shall be a maximum of 3.0w.

3.3.8 Dimensions. The outline dimensions of the transmitter shall conform to figure 2.

3.3.9 Weight. The weight of the complete fuel flowmeter transmitter shall not exceed 6.5 pounds.

3.3.10 Mounting. Mounting bosses and spacing thereof shall be as specified on figure 2.

3.3.11 Flow direction. The direction of fuel flow through the transmitter shall be permanently and legibly marked by arrow as shown on figure 2 on both front and back.

3.3.12 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. Sampling, inspection and testing of the transmitters shall be as specified in MIL-I-7566, and as specified herein. Where qualification is called out in MIL-I-7566, it shall be understood for the purpose of this specication to be first article test.

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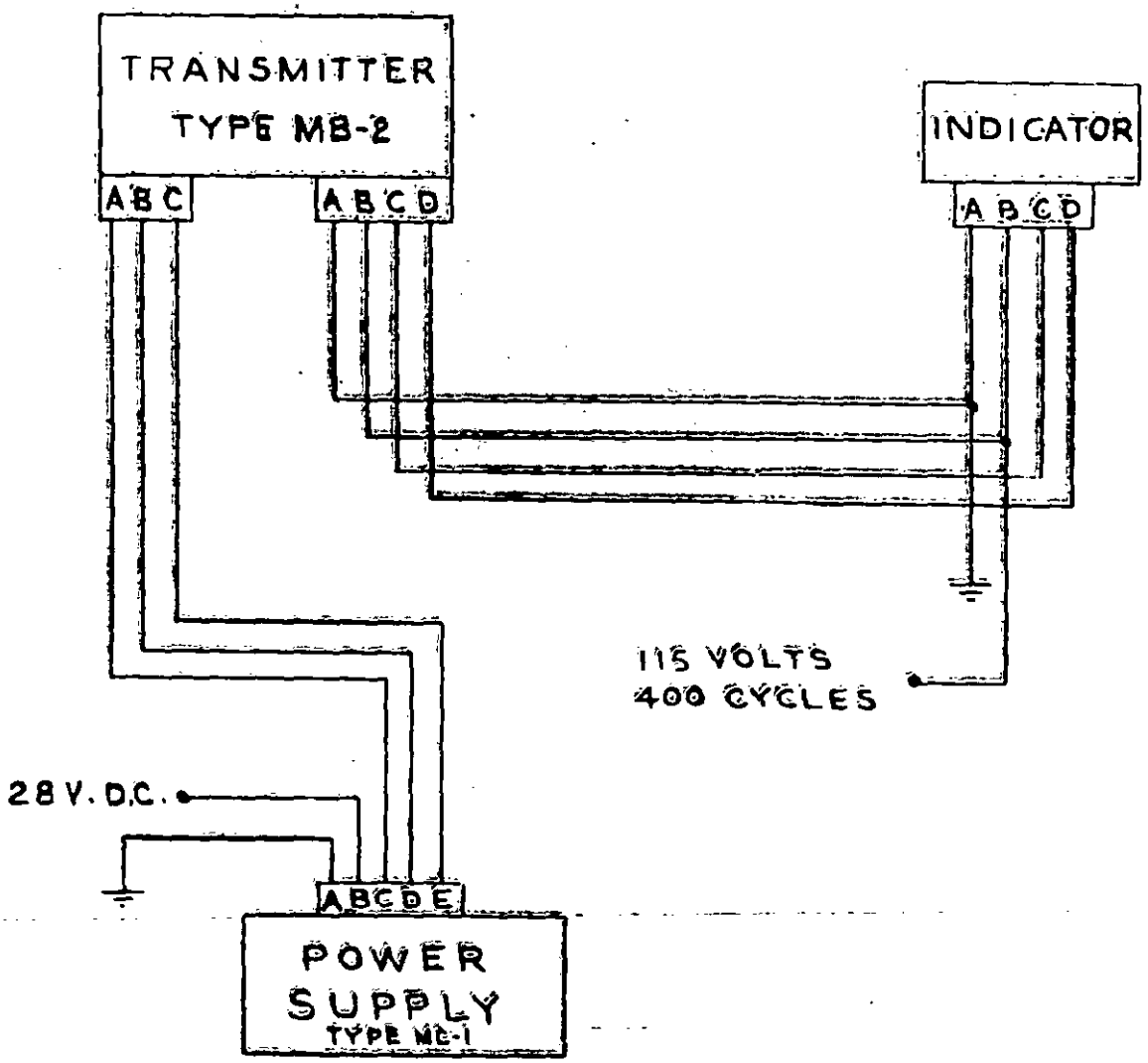
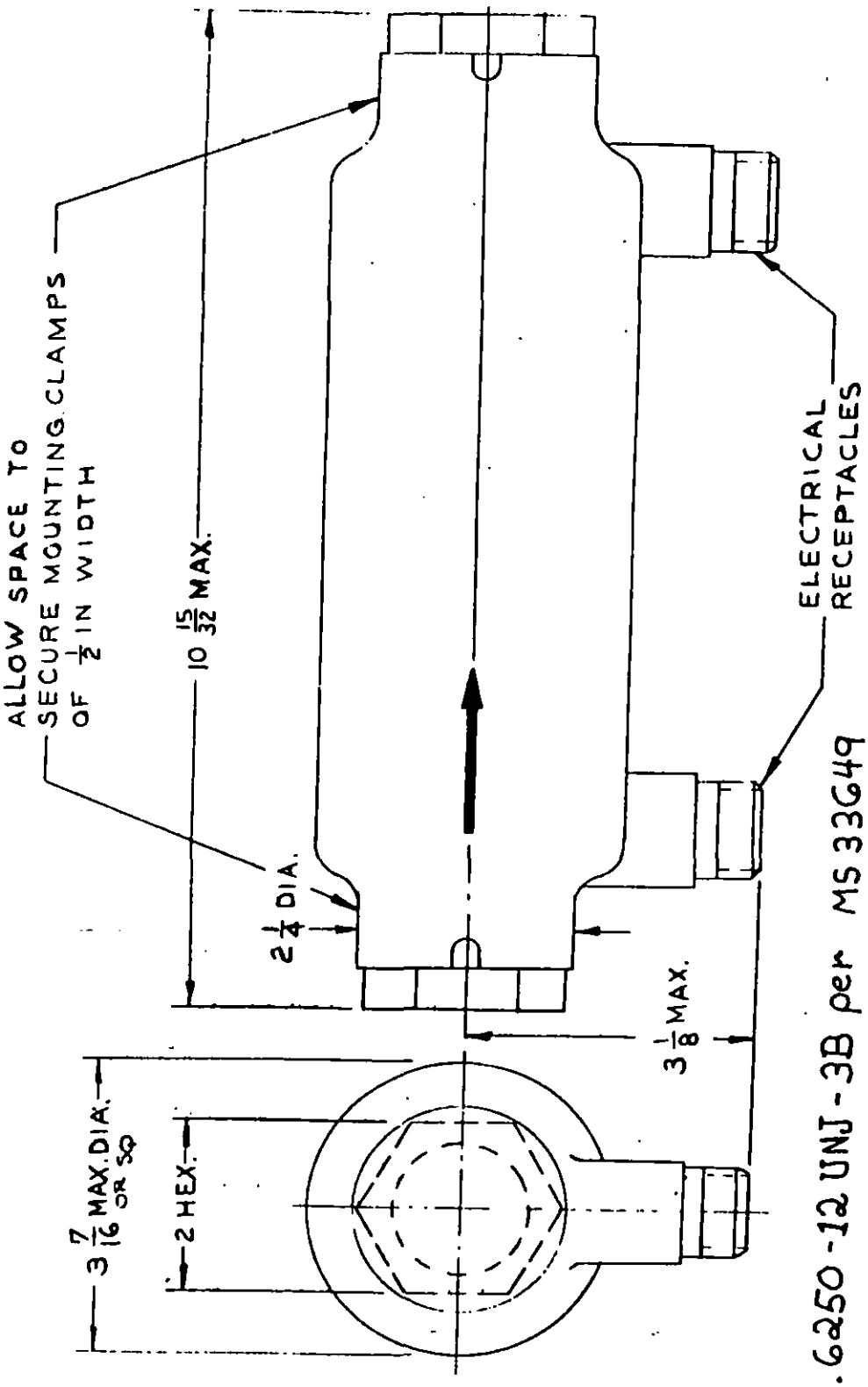


FIGURE 1. External wiring.

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DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED
TOLERANCES:
FRACTIONS: $\frac{1}{64}$

FIGURE 2 TRANSMITTER

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4.3.2.2 Low temperature exposure. The transmitter shall then be subjected to a temperature of -65°C ±2°C for a period of not less than 24 hours, and then operated at this temperature over the scale range to determine that no component parts have been damaged. The temperature shall then be raised to -54°C ±1°C for a period of not less than 24 hours. The scale errors and friction shall not exceed the low temperature tolerances as shown in tables I and II. The transmitter shall be subjected to the room temperature scale error test not less than 1 hour following the cold exposure test. The room temperature tolerances, as specified in tables I and II, shall apply.

4.3.2.3 High temperature exposure. The transmitter shall be subjected to an ambient temperature of 121°C ±2°C for a period of 4 hours with fuel in the instrument, but not flowing. During the last 30 minutes of this period, fuel at 38°C ±2°C shall be pumped through the transmitter at a rate of 1,000 pounds per hour. The transmitter shall then be tested for scale errors and friction under the same conditions. The scale errors and friction shall not exceed the tolerances specified in tables I and II.

4.3.2.4 Endurance. The scale errors after endurance test shall not exceed the tolerances specified in table I for the room temperature scale error test by more than 20 pph in the lower scale range, and 90 pph in the upper range. Friction errors shall be within the tolerances specified in table II.

4.3.2.4.1 Endurance duration. In the event the elapsed time of the endurance test is less than 1200 hours when the specified number of flow applications has been completed, then the system shall be operated at 80 ±5 percent of the range of the system until a total of 1200 hours of operation has been completed.

4.3.2.5 Contaminated fuel operation. Test fuel containing dirt to the extent of 80 grams of foreign matter to 1000 gallons of fuel shall be circulated through the transmitter at approximately 2000 pph for one hour. A scale and friction error test shall then be made with the contaminated fuel and operation shall be satisfactory and within the room temperature tolerances. This foreign matter shall be considered to consist of not less than 68 percent SiO_2 and shall have a particle-size analysis as shown in table IV.

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TABLE IV

Foreign Particle Size, Microns	Percent of Total Weight
0 - 5	39 ±2
5 - 10	18 ±3
10 - 20	16 ±3
20 - 40	18 ±3
over 40	9 ±3
Through a 200-mesh screen	100 Percent

4.3.2.6 Response time. The fuel flow shall be changed from 0 to 6000 pph by means of a rapid action cut-off valve in the fuel line. The time for the indication of a master indicator connected to the transmitter to come within one percent of its steady state position shall be within 5 seconds.

4.3.2.7 Shock test. The test shall be in accordance with procedure I of MIL-E-5272. The transmitter shall be subjected to impact shocks of 15g, each shock impulse having a time duration of 7 ±2 milliseconds. Three shocks shall be applied in each of the following directions:

a. Along the longitudinal axis.

b. At 45° to the longitudinal axis in one direction.

4.4 Inspection and test. Test methods of preservation and packaging shall be accomplished with section 4 of MIL-P-116, to insure compliance with section 5 of this specification.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be leval A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Transmitter, rate of flow, shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Transmitter, rate of flow, shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Not applicable.

5.1.1.4 Unit packaging. Unless otherwise specified by the procuring activity, each transmitter, rate of flow, shall be paekaged in quantity unit packs of one each in accordance with

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method 1C-1 of MIL-P-116. Overbox in PPP-B-636 carton large enough to allow for application of sufficient cushioning material between container and box, of a type, density and thickness to insure shock transmission does not exceed peak values in Gs established for the item, when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Transmitter, rate of flow, shall be clean, dry and individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Transmitter, rate of flow, packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, weather-resistant class, unless otherwise specified by the procuring activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, of minimum tare and cube consistent with the protection required. Table II of MIL-STD-794 applies for Army use.

5.2.2 Level B. Transmitter, rate of flow, packaged as specified in 5.1.1 shall be packed in domestic class exterior shipping containers conforming to PPP-B-636. Other requirements as specified in 5.2.1 above shall apply.

5.2.3 Level C. Packaging shall be applied which affords adequate protection during domestic shipment from the supply source to the first domestic receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the contractor's commercial practice provided the latter meets the requirements of this level.

5.3 Mark for shipment. Interior packaging and exterior shipping container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The type MB-2 fuel flowmeter transmitter covered by this specification is intended to indicate the rate of flow of fuel in pounds per hour being consumed by an aircraft jet engine when used in conjunction with a suitable indicator and power supply.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Whether sampling plan A or B is to be used.

c. Packaging (see section 5).

d. First article (see 3.1 and 4.1).

6.3 Reclaimed material. The use of reclaimed materials shall be encouraged to the maximum extent possible.

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TRANSMITTER, RATE OF FLOW, FUEL, TYPE MB-2

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